ESERCIZIO 3

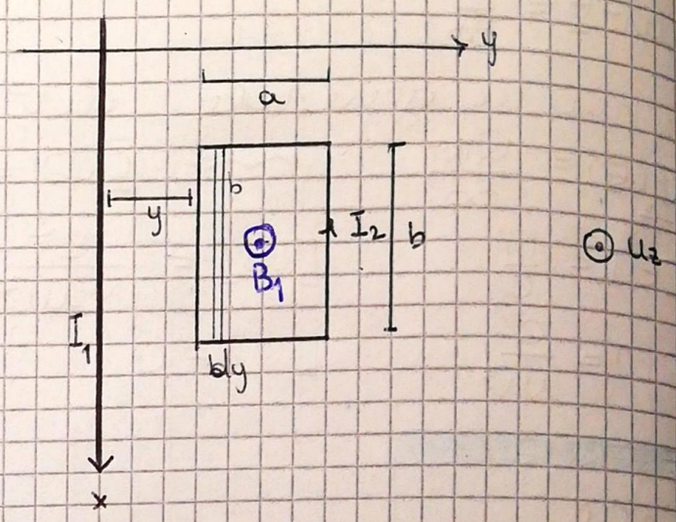
$$\underline{B}_1 = \frac{\mu_0 I_1}{2\pi y'} u_z$$

$$\Phi_{21}(\underline{B}_1) = \int_y^{y+a} \frac{\mu_0 I_1}{2\pi y'} b\,dy' =$$

$$= \frac{\mu_0 I_1 b}{2\pi} \ln\left(\frac{y+a}{y}\right) = M I_1$$

$$\Rightarrow M = \frac{\mu_0 b}{2\pi} \ln\left(\frac{y+a}{y}\right)$$

$$U = M I_1 I_2 = \frac{\mu_0 b}{2\pi} I_1 I_2 \ln\left(\frac{y+a}{y}\right)$$

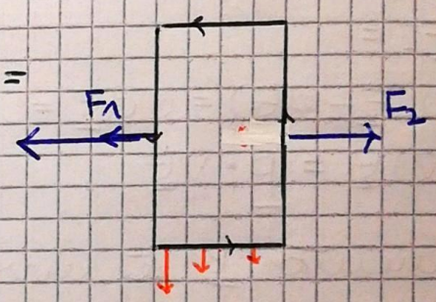


$$\underline{F}_{TOT} = \underline{F}_1 + \underline{F}_2 =$$

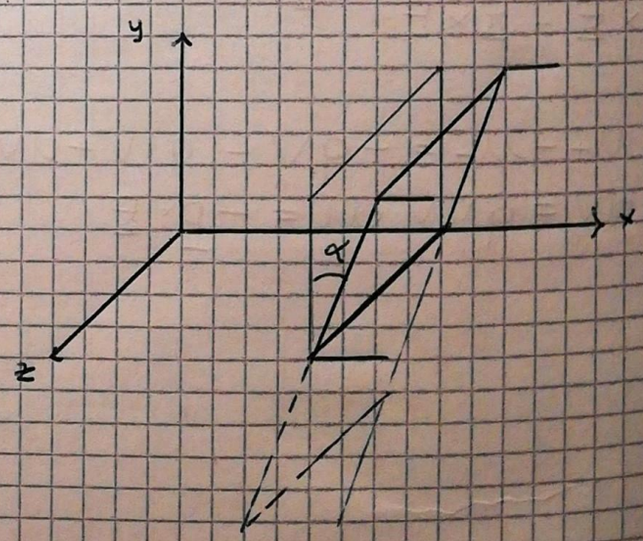
$$= B(y'=y) I_2 b\, \underline{u}_y - B(y'=y+a) I_2 b\, \underline{u}_y =$$

$$= \frac{\mu_0 I_1 I_2 b}{2\pi}\left(\frac{1}{y} - \frac{1}{y+a}\right) =$$

$$= \frac{\mu_0 I_1 I_2 b a}{2\pi y (y+a)} \underline{u}_y$$



ESERCIZIO 4

$$\underline{E} = E_0 \cos(\omega t - kx)\, \underline{u}_y$$

$$\bar{I} = \frac{1}{2} E_0^2 \sqrt{\frac{\varepsilon_0}{\mu_0}} \cos\alpha$$